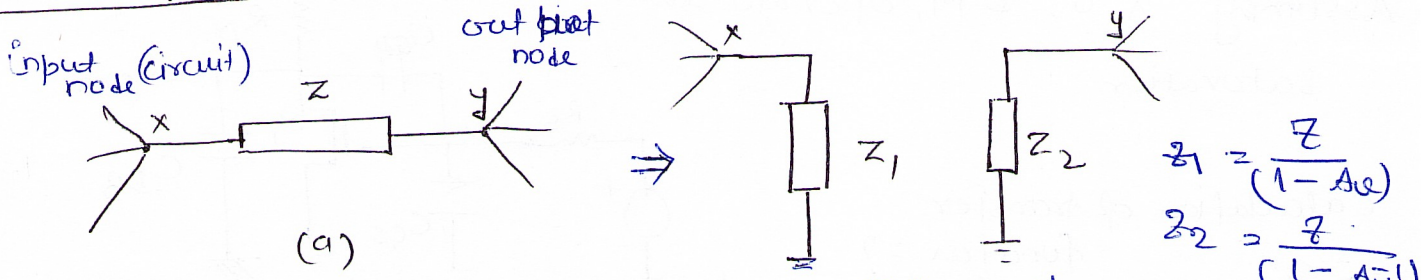
Chapter 6:- Frequency Response of Amplifier

Miller Effect:- → An important phenomenon that occurs in many analog (and digital) circuits is related to a "Miller Effect", as described by Miller in a theor



$$Z_1 = \frac{Z}{(1 - A_V)}$$

$$Z_2 = \frac{Z}{(1 - A_V^{-1})}$$

(b). where $A_V = V_y / V_x$.

proof:- → from fig (a).

Current flowing through z from x to Y $= \left(\frac{V_x - V_y}{Z}\right)$

for above circuit (a) & (b), to be equivalent, the same current must flow through Z_1;

Thus, $$\frac{V_x - V_y}{Z} = \frac{V_x}{Z_1}$$

$$\therefore \quad Z_1 = \frac{Z}{1 - \left(\frac{V_x}{V_y}\right)} \qquad \left(\frac{V_x}{V_y} = A_V\right)$$

$$\boxed{Z_1 = \frac{Z}{(1 - A_V)}}$$

similarly we can writ,

$$\boxed{Z_2 = \frac{Z}{1 - (1/A_V)}}$$

or $$Z_2 = \frac{Z}{(1 - A_V^{-1})}$$

Common-Source Stage:- The common-source topology exhibits a relatively high input impedance while providing voltage gain.

Assuming $\lambda = 0$, & M_1 operate in saturation

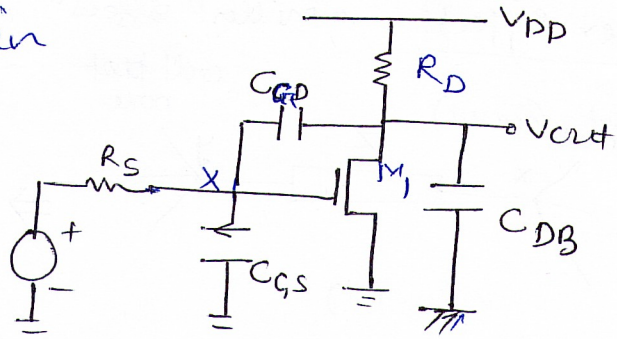


Fig:- High frequency model of CS-stage.

Calculation of transfer function =?

Estimate the transfer function by associating one pole with each node.

Total capacitance seen from X to ground,

$$C_{GS} + \underbrace{(1 - A_V) C_{GD}}_{\text{from miller effect}}$$

where, $A_V = -g_m R_D$ (Low freq. gain)

$\therefore$ magnitude of i/p pole,

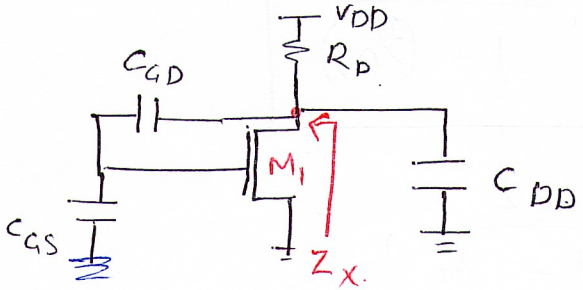
$$\omega_{in} = \frac{1}{R_{eq} \cdot C_{eq}}$$

$$\omega_{in} = \frac{1}{R_S \left[C_{GS} + (1 + g_m R_D) C_{GD}\right]} \quad \text{---(1)}$$

Calculation of Output pole =?

* At the o/p node, the total capacitance seen to ground is equal to C_{DB} plus the miller effect of C_{GD}: $C_{DB} + (1 - A_V) C_{GD} \approx C_{DB} + C_{GD}$

R_S is relatively large then above circuit simplified as,



effect of R_S neglected.

** $\omega_{out} = \frac{1}{R_D (C_{DB} + C_{GD})}$

$$\therefore \quad Z_x = \frac{1}{C_{eq} s} \cdot \| \left(\frac{C_{GD} + C_{GS}}{C_{GD}} \right) \cdot \frac{1}{g_m}$$

where $$C_{eq} = \left(\frac{C_{GD} C_{GS}}{C_{GD} + C_{GS}} \right)$$

$\therefore$ output pole approximately, equal to,

$$\omega_{out} = \frac{1}{R_{eq} \cdot C_{eq}}$$

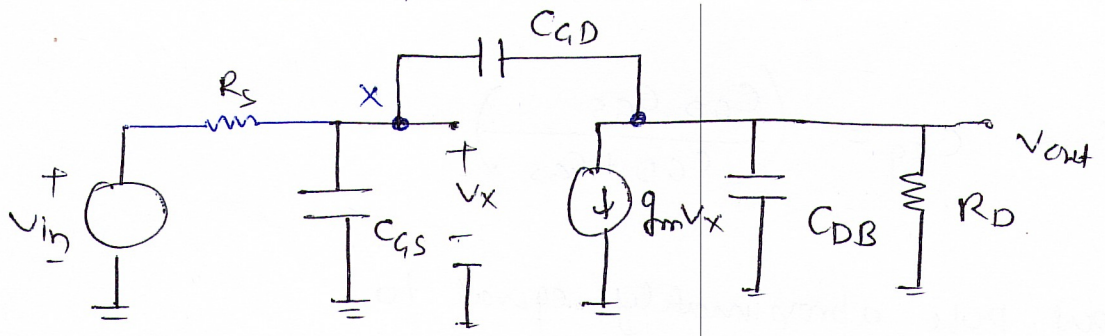
$$= \frac{1}{\left[R_D \| \left(\frac{C_{GD} + C_{GS}}{C_{GD}} \cdot \frac{1}{g_m} \right) \right] (C_{eq} + C_{DB})} \quad - (2)$$

Hence we can write transfer function,

$$\frac{V_{out}}{V_{in}}(s) = \frac{-g_m R_D}{\left(1 + \frac{s}{\omega_{in}}\right)\left(1 + \frac{s}{\omega_{out}}\right)}$$

In above calculation we have not considered the existence of zeros in the circuit. And another concern stem from approximating the gain of the amplifier by $-g_m R_D$ whereas in reality the gain varies with frequency.

now to calculate the exact transfer function - consider following equivalent circuit,



using KCL at above circuit,

$$\frac{V_X - V_{in}}{R_S} + V_X C_{GS} s + (V_X - V_{out}) C_{GD} s = 0 \quad - (3)$$

&

$$(V_{out} - V_X) C_{GD} s + g_m V_X + V_{out}\left(\frac{1}{R_D} + C_{DB} s\right) = 0 \quad - (4)$$

from equation (4)

$$V_X = \frac{V_{out}\left(C_{GD} s + 1/R_D + C_{DB} s\right)}{g_m - C_{GD} s} \quad - (5)$$

put eq (5) in eq (3)

$$\frac{-V_{out}\left[1/R_S + (C_{GS} + C_{GD}) s\right]\left[\frac{1}{R_D} + (C_{GD} + C_{DB}) s\right]}{g_m - C_{GD} s} - V_{out} C_{GD} s = \frac{V_{in}}{R_S}$$

while solving above equation,

$$\frac{V_{out}}{V_{in}}(S) = \frac{(C_{GD}S - g_m) R_D}{R_S R_D \xi S^2 + [R_S(1+g_m R_D) C_{GD} + R_S C_{GS} + R_D(C_{GD}+C_{DB})]S + 1}$$

where

$$\xi = C_{GS}C_{GD} + C_{GS}C_{DB} + C_{GD}C_{DB}$$

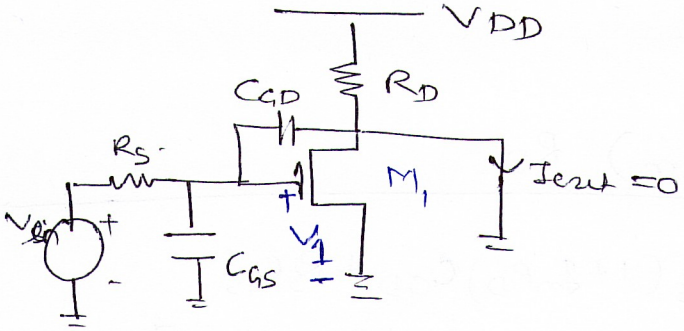
Note:- The Transfer function is of 2nd order even though the circuit contain three capacitor. This is because the capacitors form a loop, allowing only two independent initial condition in the circuit hence yielding 2nd order D.E. for the time response.

Calculation of zeros in a CS stage:

→ The S_Z can also be calculated by noting that the T.F. $\frac{V_{out}(S)}{V_{in}(S)}$ must drop to zero for $S = S_Z$.

→ for finite V_{in}, this mean, $V_{out}(S_Z) = 0$

⇒ output can be shorted to ground, at this frequency with no current flow through the short as shown below in following circuit,



The current through C_{GD} and M_1 are equal and opposit,

$$V_1 C_{GD} s_z = g_m V_1$$

$$\therefore \quad \boxed{s_z = \frac{g_m}{C_{GD}}}$$

Calculation of Input impedence:-

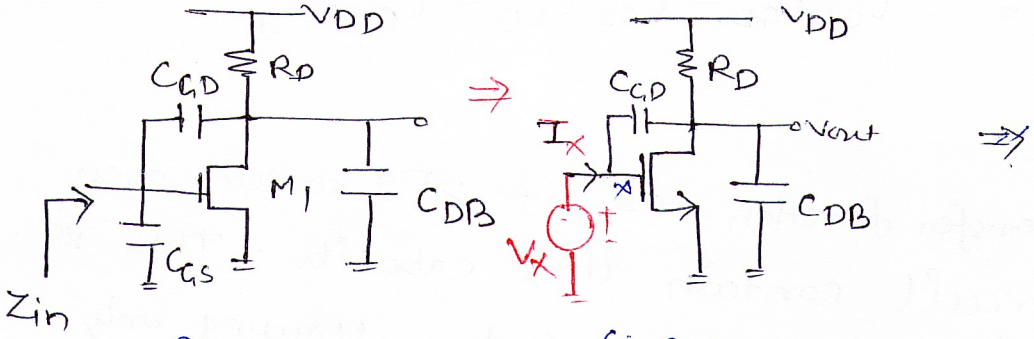


fig.(a) fig(b)

As first order approximation, from fig(a)

$$\boxed{Z_{in} = \frac{1}{[C_{GS} + (1 + g_m R_D) C_{GD}]\, s}}$$

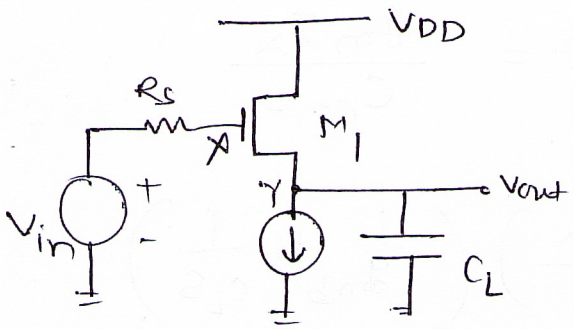
But at high freq. the effect of output node must be taken into account. Ignoring C_{GS} for moment and using ~~circ~~ circuit of fig(b)

$$(I_X - g_m V_X) \frac{R_D}{1 + R_D C_{DB} s} + \frac{I_X}{C_{GD} s} = V_X$$

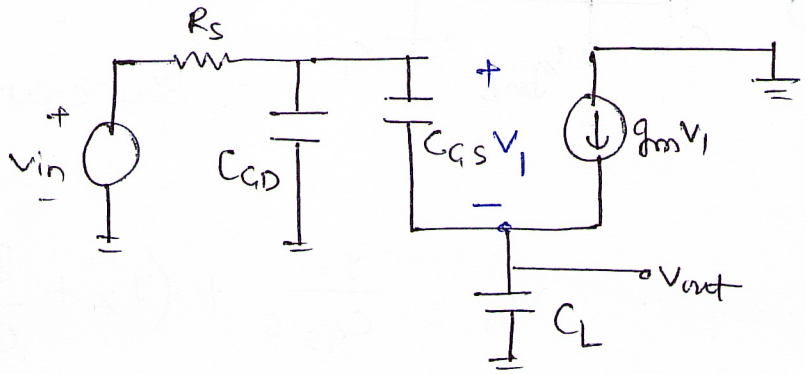
$$\therefore \quad \boxed{Z_{in} = \frac{V_X}{I_X} = \frac{1 + R_D (C_{GD} + C_{DB}) s}{C_{GD} s [1 + g_m R_D + R_D C_{DB} s]}}$$

Source follower :-

→ Source follower are occasionally employed as level shifter or buffers, impacting the overall frequency response.



(a) Source follower



(b) high freq. equivalent circuit

from above circuit (b) we can sum the currents at output node:

i.e. $$V_1 C_{GS} s + g_m V_1 = V_{out} C_L s \quad -(1)$$

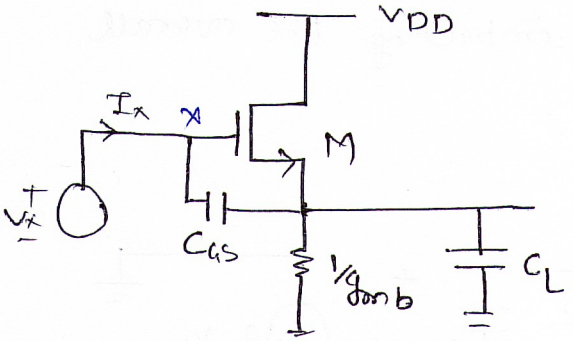
$$\therefore \quad V_1 = \frac{C_L s}{g_m + C_{GS} s} \cdot V_{out} \quad -(2)$$

& $$V_{in} = R_S \left[V_1 C_{GS} s + (V_1 + V_{out}) C_{GD} s \right] + V_1 + V_{out} \quad -(3)$$

from (2) & (3)

$$\boxed{\frac{V_{out}}{V_{in}}(s) = \frac{g_m + C_{GS} \cdot s}{R_S (C_{GS} \cdot C_L + C_{GS} C_{GD} + C_{GD} C_L) s^2 + (g_m R_S C_{GD} + C_L + C_{GS}) s + g_m}}$$

Calculation of i/p impedance :-



→ C_{GD} simply shunt the i/p and can be ingnored

V_{GS} of M_1 = $\frac{I_x}{C_{GS} \cdot s}$

∴ Source current = $\frac{g_m I_x}{C_{GS} \cdot s}$

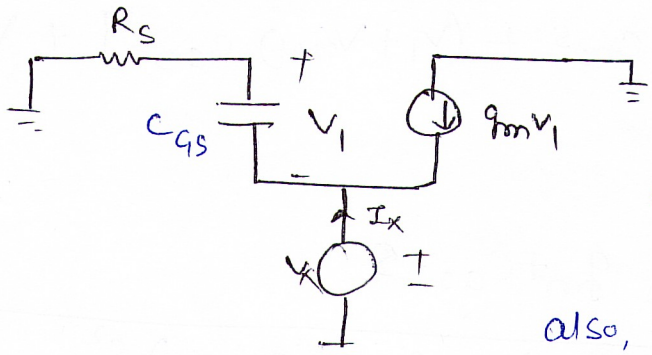
$$V_x = \frac{I_x}{C_{GS} \cdot s} + \left(I_x + \frac{g_m I_x}{C_{GS}\, s}\right)\left(\frac{1}{g_{mb}} \,||\, \frac{1}{C_L s}\right)$$

$$\therefore \boxed{Z_{in} = \frac{V_x}{I_x} = \frac{1}{C_{GS} s} + \left(1 + \frac{g_m}{C_{GS} s}\right) \frac{1}{g_{mb} + C_L s}}$$

at relatively low freq. $g_{mb} \gg |C_L s|$

$$\therefore \quad Z_{in} \simeq \frac{1}{C_{GS}\, s} \cdot \left(1 + \frac{g_m}{g_{mb}}\right) + \frac{1}{g_{mb}}$$

Calculation of output impedance :-



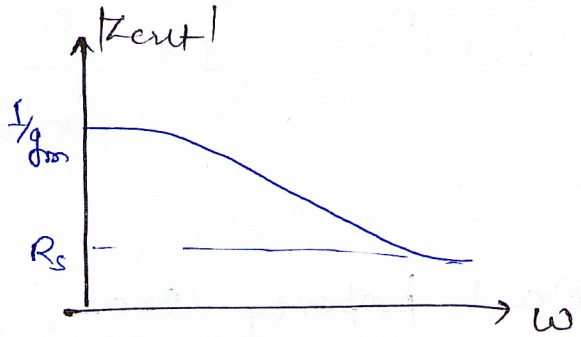
$$V_1 C_{GS} \cdot s + g_m V_1 = -I_x$$

also, $V_1 C_{GS} s R_S + V_1 = -V_x$.

$$Z_{out} = \frac{V_x}{I_x} = \frac{R_S C_{GS} s + 1}{g_m + C_{GS} s}$$

at low freq. $Z_{out} \simeq 1/g_m$

at high freq. $Z_{out} \simeq R_S$ (because C_{GS} short the gate and source)



3) Common Gate Stage:-

In common gate stage the i/p and o/p nodes are isolated if channel length modulation is neglected.

Calculation of Transfer function:-

similar to above calculation of CS, we can get,

$$V_{in} = \frac{V_{out}}{} \cdot s$$

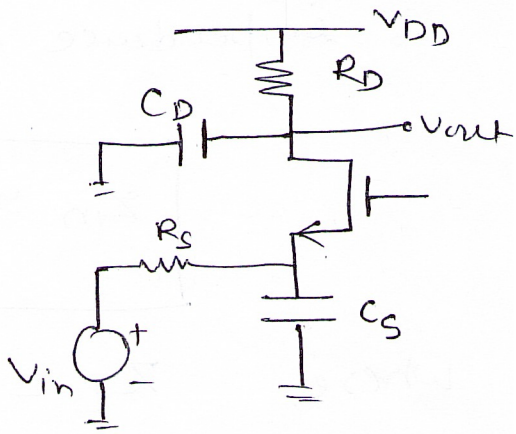


Fig:- Common gate Stage at high frequency

$$\frac{V_{out}}{V_{in}}(s) = \frac{(g_m + g_{mb}) R_D}{1 + (g_m + g_{mb}) R_S} \cdot \frac{1}{\left(1 + \frac{s}{\omega_{in}}\right)\left(1 + \frac{s}{\omega_{out}}\right)}$$

$$\left[\frac{V_{out}}{V_{in}}(s) = \frac{(g_m + g_{mb}) R_D}{1 + (g_m + g_{mb}) R_S} \times \frac{1}{\left(1 + \frac{C_S \cdot s}{g_m + g_{mb} + R_S^{-1}}\right)(1 + R_D C_D s)}\right]$$

→ CG has important property that, it exhibit no miller multiplication of capacitances, potentially achieving wide band

Input impedance of CG topology does depend on the drain load if $\lambda \neq 0$

∴ impendance seen looking into the source of M_1,

$$\boxed{Z_{in} \simeq \frac{Z_L}{(g_m + g_{mb}) r_o} + \frac{1}{(g_m + g_{mb})}}$$

where, $Z_L = R_D \,||\, [1/C_D s]$ ⇒ Z_{in} depend on Z_L

Cascode Stage:-

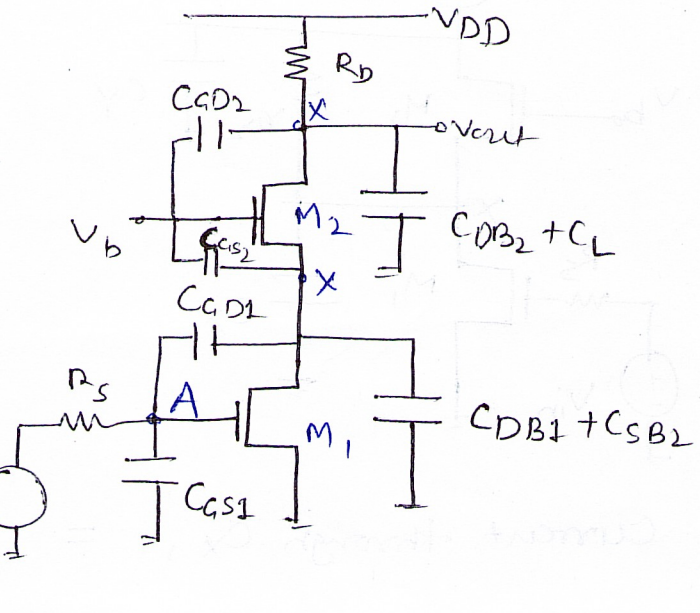
A cascade of a common-source stage and common gate stage, a cascode circuit offers the speed of the latter - by supressing the miller effect and the i/p impedence of the former.

@ node A from circuit,

pole associated with node 'A'

$$\omega_{P,A} = \frac{1}{R_{eq} \cdot C_{eq}}$$

$$= \frac{1}{R_S\left[C_{GS_1} + \left(1 + \frac{g_{m_1}}{g_{m_1} + g_{mb_2}}\right) C_{GD_1}\right]}$$



@ node 'X' total capacitence $= 2C_{GD_1} + C_{DB_1} + C_{SB_2} + C_{GS_2}$

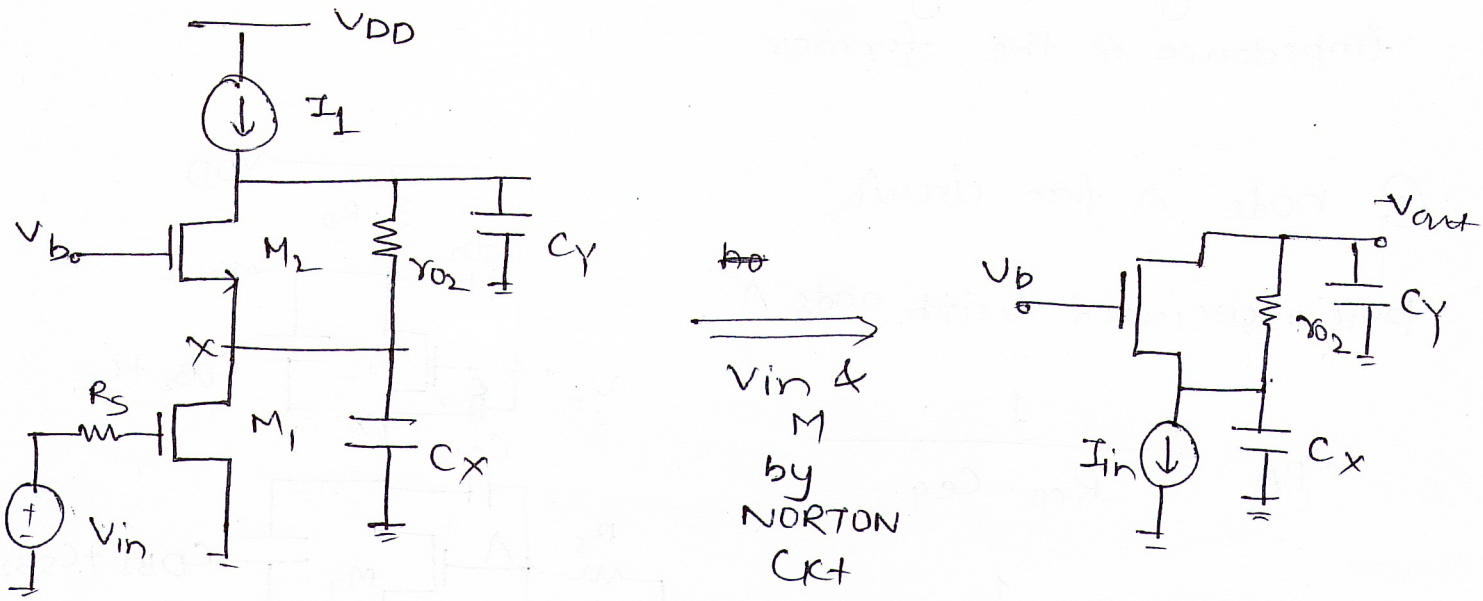
$$\therefore \omega_{P,X} = \frac{(g_{m_2} + g_{mb_2})}{\left[2C_{GD_1} + C_{DB_1} + C_{SB_2} + C_{GS_2}\right]}$$

@ output node 'Y',

$$\omega_{P,Y} = \frac{1}{R_D\left(C_{DB_2} + C_L + C_{GD_2}\right)}$$

now, 1) in above circuit replace load resistor with current source

→ 2) Neglecting capacitance associated with M_1,



Current through C_X, $= -V_{out} C_Y s - I_{in}$.

& $$V_X = -\frac{(V_{out} C_Y s + I_{in})}{C_X s}$$

small signal drain current of M_2 $= -g_{m_2}\left(-V_{out} C_Y s - I_{in}\right) / (C_X s)$

Current through r_{o_2} $= -V_{out} C_Y s - g_{m_2}(V_{out} C_Y s + I_{in}) / (C_X s)$

V_X + voltage drop across r_{o_2} = V_{out}

$$\therefore \quad -r_{o2}\left[(V_{out}C_Y s + I_{in})\frac{g_{m_2}}{C_X s} + V_{out}C_Y s\right] - (V_{out}C_Y s + I_{in})\frac{1}{C_X s} = V_{out}$$

$$\therefore \quad \frac{V_{out}}{I_{in}} = \frac{-g_{m_2}r_{o2}+1}{C_X s} \cdot \frac{1}{(1+g_{m_2}r_{o2})\frac{C_Y}{C_X} + C_Y r_{o2} s}$$

for $g_{m_2}r_{o2} \gg 1$ & $g_{m_2}r_{o2}C_Y/C_X \gg 1$ i.e. $C_Y > C_X$

$$\therefore \quad \frac{V_{out}}{I_{in}} = \frac{-g_{m_2}}{C_X s}\left[\frac{1}{\frac{C_Y}{C_X}g_{m_2} + C_Y s}\right]$$

hence, $$\left[\frac{V_{out}}{V_{in}}(s) = \frac{-g_{m_1}g_{m_2}}{C_Y C_X s} \cdot \frac{1}{(g_{m_2}/C_X + s)}\right]$$